Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Dr. Cornelius Dirksen	Email	cornelius.dirksen@ikts.fraunhofer.de
Institute Name	Fraunhofer IKTS			
Address	Michael-Faraday-Str. 1, 07629 Hermsdorf, Germany			

2. Technology Overview

Title of Proposed Technology	Fluorine-free gas diffusion electrodes for the application in alkaline electrolysis, fuel cells and metal-air batteries			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Energy & Resources		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	Hydrogen	Polymeric Materials	Materials for Energy/Environmental Industries
Concept of Proposed Technology	- Fluorinated polymers are among the forever chemicals and are therefore difficult to break down - PFAS-free polymers are a solution for a broad kind of processes, e.g. hydrogen production processes, fuel cells, GDEs for metal-air batteries or CO_2 electrolysis - Usage of fluorine-free polymers in GDEs to prevent additional waste, microplastics and have more sustainable materials for a green environmental-friendly future			
Rationale for Joint R&D Project	- Significant reduction in development time and costs is possible through overseas collaboration - International cooperation is necessary for fields aimed at solving global challenges such as climate change			

	- Joint testing under different conditions, operating philosophies and regulatory frameworks (EU and Korea) ensures that the new GDEs are globally deployable - Shared IP and joint development reduce risk and accelerate time-to-market for companies in both regions, opening access to both European and Asian markets - No single country covers the full chain from materials discovery → GDE design → cell/stack integration → industrial validation → scale-up at competitive cost
Key Words	1. *Hydrogen production* 2. *Fluorine-free* 3. *Gas diffusion electrode*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)

R&D Objectives

- Prototype of a fluorine free gas diffusion electrode. Avoidance of e.g. Teflon® to achieve hydrophobic behavior of the gas diffusion electrode, but utilizing a new fluorine free polymer system
- proof of concept utilizing an alkaline electrolysis apparatus in relevant size, current density and run time
- proving scale up strategies for polymer and electrode production

R&D Target Specifications

- Fluor free polymer for gas diffusion electrodes
- Electrode design utilizing the new polymer system
- scale up strategies for electrode production and polymer production
- Develop and validate PFAS-free gas diffusion electrodes for applications like alkaline electrolysis with:
  - comparable or better performance (≥ 0.5–1.0 A/cm², low overpotential),
  - lifetime ≥ 20,000 h,
  - stability in 30–40 wt-% KOH at 60–90 °C
  - industrially scalable manufacturing (coating processes)

Geometric / material-related parameters

- total GDE thickness: ≤ 500 µm
- porosity for gas transport: > 60 %
- hydrophobic agent content (PFAS-free): ≥ 5%

Electrochemical performance parameters

- contact angle (water): ≥ 90°
- Overpotential at nominal current density: 50 – 400 mV
- Operating current density: ≥ 0.5 A/cm²
- Exchange current density (HER/OER): 10^{-7} – 10^{-3} A/cm²

R&D Plan

Key tasks / methodologies:

- Synthesis of block copolymers Block copoly(phenylquinoxaline)s, surface functionalization and optimization
- Advanced materials analysis, modeling of gas/liquid transport, degradation mechanisms
- Develop fluorine-free binders, hydrophobizing agents, catalysts
- testing under realistic conditions

Expected Milestones / Timeline:

- Milestone 1 / Year 1: PFAS-free hydrophobizing agent evaluate and synthesized
- Milestone 2 / Year 2: fully working GDE with new binder and agents
- Milestone 3 / Year 3: demonstration of GDE in alkaline demo cell

Workplan:

WP	Title	Short content description
1	Requirements & design rules	Define specs (current density, temperature, pressure, lifetime); analyze state-of-the-art PFAS-based GDEs; derive material and structural requirements specifically for AEL.
2	Material development	Screening/synthesis of fluorine-free binders;

		(PFAS-free)	hydrophobizing agents; compatible catalysts (Ni systems).
	3	GDE design & manufacturing	Develop layer architectures (gas diffusion layer + catalyst layer); vary porosity, layer thickness, binder content; evaluate manufacturing routes (galvanization).
	4	Electrochemical characterization	Half-cell and single-cell tests in KOH: polarization curves, EIS, limiting current densities, Faradaic efficiency; comparison with PFAS-containing reference GDEs.
	5	Demonstrator & integration	Implement best PFAS-free GDEs in an AEL demo cell/stack; operation under realistic conditions; validation of scalability and operational safety.

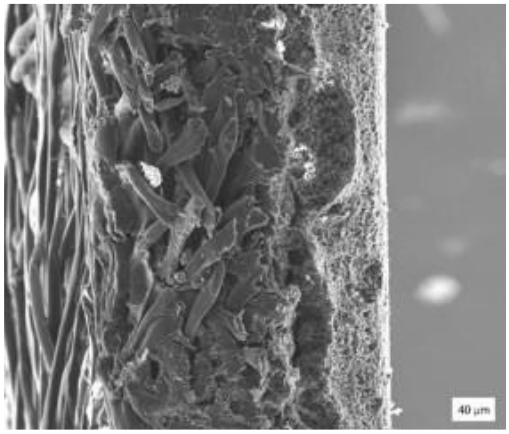
4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute

Fraunhofer IKTS:

- Design and scale up of the gas diffusion electrode utilizing the novel polymer system
- scale up experience up to 20 kW demonstrators; different characterizing methods
- Testing the gas diffusion electrode in an alkaline an alkaline electrolysis apparatus in relevant size, current density and run time
- Transfer of knowledge and process parameters to Korean partners

Example of a GDE from IKTS:



Fraunhofer IAP:

- Expertise of over 20 years in organic chemistry – synthesis of tailored molecules, monomers and polymers
- Design and scale up strategies (from lab to industry scale) for the novel PFAS-free ionic conductive polymer systems
- Testing and analyzing polymer characteristics
- Transfer of knowledge and synthesis technology to Korean partners

Synthesis of a PFAS-free ionic conductive membrane polymers

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Outcome: ○ Fluor free polymer material for gas diffusion electrodes including process parameters and operating parameters ○ Scale up strategies for polymer and gas diffusion electrode production (including cost models) ○ Alkaline electrolysis apparatus utilizing the novel gas diffusion electrode ○ Basis for industrialization Impact: ○ Fluorine-free polymers in electrolysis have the potential to significantly reduce PFAS-related environmental and health risks, simplify disposal, and reduce regulatory risks ○ Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.

6. References

(1) Bekisch, A.; Skadell, K.; Ast, J.; Schulz, M.; Weidl, R.; Christiansen, S.; Stelter, M.; (2023). Influence of pore size on mass transport: Bifunctional MnOx-coated nickel foam vs carbon corrosion prone commercial GDE in alkaline electrolyte. International Journal of Hydrogen Energy. 50. 10.1016/j.ijhydene.2023.07.251.

(2) Bekisch, A.; Skadell, K.; Ast, J.; Schulz, M.; Weidl, R.; Christiansen, S.; Stelter, M. ;(2025). Alternative Gas Diffusion Electrode Designs: Influence of Porosity Gradients on the Electrochemical Activity. Advanced Energy and Sustainability Research. 6. 10.1002/aesr.202400202.

(3) Janietz, S.; Egorov-Brening, T., Schiestel, Th.; Götz T.; (2023). Block copoly(phenylquinoxaline)s as potential ionomers for proton exchange membranes. Polymer. Volume 271. 10.1016/j.polymer.2023.125794.

(4) Janietz, S.; Egorov-Brening, T.; Krüger, H.; (2017). Verfahren zur Herstellung von Blockcopolychinoxalinen mit Sulfon- oder Phosphonsäure-Gruppen und deren Einsatz als protonenleitende Membran (DE102017208805B4); Deutsches Patent- und Markenamt.

(5) Janietz, S.; Egorov-Brening, T.; (2019). Quarternäres Polychinoxalin, Verfahren zu dessen Herstellung, Hydroxidionen leitende Membran und Verfahren zu deren Herstellung (DE102019203597B4); Deutsches Patent- und Markenamt.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization